

The Insurance Value of Public Insurance Against Idiosyncratic Income Risk^{*}

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Abstract

The tax and transfer system partially insures households against individual income risk. We discuss under which assumptions differences between income processes estimated for household gross income and disposable income are informative about the (welfare) value of this partial insurance. Our approach works directly with income processes estimated separately on the two income measures, and does not require the specification (nor estimation) of a tax function. Instead we use an incomplete markets framework that links an estimated income process to consumption. Its key feature is that the degree of partial insurance is directly parameterized: Technically, this allows to solve for the degree of insurance provided by the tax and transfer system as a fixed point. The approach works with standard restrictions on income processes and preferences, and it further enables us to explore the role of higher-order risk for the value assigned to public insurance.

Keywords: Idiosyncratic income risk, tax and transfer system, social insurance policy, incomplete markets.

1 Introduction

Motivation. Most individuals face individual income risk: firm closures, unemployment spells, or health shocks are just some of the sources of downside risk, which together with typical career ups and downs lead to volatile income trajectories. The institutional framework in many economies offers a mix of various policy instruments that jointly buffer disposable

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income against this volatility. Some of these policies are explicitly designed to buffer specific sources of downside risk: most notably, the unemployment insurance system dampens temporary income losses due to job loss. Similarly, disability insurance insures against severe health risk. Progressivity of the income tax system implies insurance from an ex ante perspective as it compresses the distribution of possible income changes, without targeting explicit reasons for income losses. Existing evidence for various countries documents that, on average, the overall tax and transfer schemes, in tandem with private insurance mechanisms, are successful in providing partial insurance against individual risk (see, e.g., [Blundell *et al.*, 2014](#); [De Nardi *et al.*, 2021](#)).

Question. In this paper, we assess the *insurance value* of the existing tax and transfer system. In particular, we explore this in a setting where data is available on household-level incomes before and after taxes and transfers. This is a typical setting as many rich (administrative) data sources allow for the detailed exploration of household income trajectories, but do not cover equally reliable data on consumption or consumption expenditures. Our approach rests on a flexible analytical model framework, which allows us to parameterize the degree of partial insurance. Within our framework, we can flexibly account for three features of individual income risk: First, income risk is in part transitory and in part persistent. Second, income risk is distributed asymmetrically, with positive changes and negative changes of the same magnitude not being equally likely. Third, individual income risk changes systematically over the business cycle.

Overview. The first step of the analysis is thus the formulation of a statistical income process that captures these three data aspects. First, the separation of transitory and permanent income changes. This is relevant as transitory and permanent income changes are substantially different in nature, and while transitory risk can typically be insured against better through private channels (like savings), permanent risk cannot be smoothed away at the level of households. Second and third, the distribution of individual income and its cyclicalities. Most dominantly, in times of aggregate economic downturns, individuals face a larger risk of income losses (i.e., increased downside risk) along with a smaller chance of income gains (i.e., reduced upside risk). This translates into asymmetric shifts of the distribution of individual

income changes over the business cycle—the resulting relationship between aggregate and idiosyncratic income changes is described well by procyclical skewness of the distribution of idiosyncratic changes. Such procyclical skewness is well-documented for a large set of diverse countries (see, e.g., [Busch *et al.*, 2022](#); [Guvenen *et al.*, 2014](#)). [Busch *et al.* \(2022\)](#) document that taxes and transfers dampen both the extent of income changes, downside risk, and its cyclicity—in the United States, Germany, or Sweden: household disposable income exhibits less dispersed and more symmetrically distributed changes than household gross income, and the distribution is less responsive to the aggregate business cycle. Reflecting these insights from the literature, when formulating our income process, we allow for the distribution of income shocks to vary with aggregate output.

The second and main step of the analysis is the setup of a model as a device for the measurement of the insurance value of taxes and transfers. In this context, note that consumption eventually translates into welfare. Given household-level income processes that capture the distributional regularities of gross income and net income, and given a measure of consumption, one can use some measurement device to trace out the degree of insurance coming from the tax and transfer system. Before formulating this measurement device, in the absence of consumption data we thus need to impose a model for how disposable income maps into household consumption. To this end, we employ a consumption function that takes the permanent and transitory income shocks as input, and which is parameterized by the degrees of insurance against the two types of shocks. The important feature of this consumption function is that it maps income shocks into consumption given some (exogenously fixed) degree of partial insurance. Finally, assigning welfare (and thus an insurance value) to the obtained insurance measure requires the specification of preferences.

In sum, there are three main elements of the analysis: (i.) an income process which gives a distinction between transitory and permanent components, (ii.) a consumption function which translates transitory and permanent shocks into consumption, and (iii.) a preference specification which allows for welfare interpretation. In our approach, which we sketch next, we explicitly derive the consumption function of step (ii.) from the same model that we then use to assign a welfare value in step (iii.).

Our Measurement Device. Our measurement device builds on standard incomplete market models (SIM), which feature two key ingredients: exogenous idiosyncratic productivity risk, and access to only partial insurance against it through savings in a riskfree asset (cf. [Aiyagari, 1995](#); [Huggett, 1993](#); [Imrohoroglu, 1989a](#)). Our approach is agnostic with respect to the exact source of insurance. To this end, we adopt an island model structure that features the abstraction of two distinct types of risks—one perfectly insurable, and one uninsurable (a la [Heathcote *et al.*, 2014](#), henceforth HSV). In this economy, a fraction of the overall idiosyncratic risk is shared among a group of individuals, and a fraction is purely idiosyncratic. In an equilibrium with no asset trade across islands, this implies that the fraction of total shocks attributed to within-island idiosyncratic shocks can be interpreted as the degree of partial insurance against (total) idiosyncratic risk. We use the fact that the ratio of island-level shocks to total shocks is exogenous, which makes it the key parameter of our measurement device: For a given degree of partial insurance, the (exogenous) income process maps into an (endogenous) consumption process.

We now use this solution of the model that maps the properties of idiosyncratic income risk to consumption to trace out the degree of partial insurance provided by the tax and transfer system. We consider two scenarios in the model. In the first scenario, we treat a stochastic income process that captures regularities of post-government earnings as fundamental. The model delivers a consumption process given an amount of insurance against risk remaining after taxes and transfers. In the second scenario, we then treat an income process that captures regularities of pre-government earnings as fundamental. The model now delivers a consumption process given an amount of partial insurance against risk *provided by* taxes and transfers. The idea is now to search for this degree of partial insurance as a fixed point for which the model in the second scenario (pre-government income process) yields a consumption process that makes households ex-ante indifferent to the consumption process obtained in the first scenario (post-government income process). Thus, the model serves as a measurement device for the degree of insurance coming from taxes and transfers, which takes as inputs (estimated) income processes for gross income and net income, and makes a minimal set of structural assumptions—on preferences on the degree of partial insurance beyond taxes and transfers.

Application to Sweden. We apply our approach to estimated income processes for income moments from Swedish tax data. To generate non-zero skewness and excess kurtosis consistent with the data, we specify the econometric model for (log) income as the sum of a permanent process and a transitory component, where all innovations (shocks) are drawn from mixtures of normals. We estimate two sets of parameters of this process separately for pre- and post-government household labor income by matching moments that capture the salient features of household income change distributions and their cyclical properties as documented in [Busch *et al.* \(2022\)](#).¹ We find that the degree of partial insurance provided by the tax and transfer system amounts to about 43%, which translates into a welfare gain, expressed as a consumption equivalent variation (CEV), of about 14.3% under log utility. We then focus on the part of that gain that is attributable to smoothing business cycle variation of the distribution. Taxes and transfers insure about 6% of the cyclical changes in the distribution which translates into a CEV of about 1.3%. However, the remaining risk (in post-government household-level income) is still substantial: households are willing to pay 4.6% of their consumption to completely eliminate procyclical fluctuations in skewness.

We then explicitly explore the role of taking into account higher-order risk in comparison to a Gaussian distribution that shares the same variance with the distributions considered in the benchmark analysis. We find that with log utility it does not matter much for the overall insurance gain of the tax and transfer system, which is not surprising as risk attitudes against skewness and kurtosis are relatively weak in this specification. Still, the insurance against cyclical variations in the distribution is valued twice as much under a Gaussian distribution. Thus, when not taking into account higher-order risk one would overestimate the insurance value of the tax and transfer system against cyclical variations in idiosyncratic risk. Similarly, one would overestimate the potential gain of further smoothing.

After a short literature discussion that follows, the rest of the paper is organized as follows. Section 2 builds up the measurement of partial insurance from established definitions. Section 3 outlines the quantitative model used as our measurement device. Section 4 in-

¹Note that the specific parametric form of the distribution is not essential, as long as relevant moments of the distribution are matched; see, e.g., [Busch and Ludwig \(2024\)](#), who illustrate how central moments of the distribution map into choices of agents in a life-cycle model.

roduces the income process, which disentangles transitory and permanent components of income, and serves as the input for the measurement. Section 5 discusses the quantitative results, and Section 6 concludes.

Related Literature. There is an extensive literature on the welfare benefits of tax and transfer systems across the globe. For the case of Sweden, [Floden and Linde \(2001\)](#) found large welfare gains from redistribution and insurance against uninsurable income risk. In addition, certain public insurance instruments act as automatic stabilizers against aggregate fluctuations ([McKay and Reis, 2016](#)). Drawing on recent empirical findings by [Busch *et al.* \(2022\)](#), we aim to gain insights into the welfare implications of tax and transfer systems for mitigating the pass-through of aggregate fluctuations to individual income.

Our analysis of insurance against cyclical risk links our analysis to the literature on the welfare costs of business cycles, which has a long history, tracing its origins to the pioneering work of [Lucas \(1987\)](#) but widely generalized to the context of heterogeneous agents facing idiosyncratic income risk and incomplete markets ([Imrohoroglu, 1989b](#); [Storesletten *et al.*, 2001](#); [Krusell *et al.*, 2009](#)). This literature emphasizes the role of distributions and cyclical variation in idiosyncratic income risk as a source of amplification of the welfare costs of cyclical fluctuations. The distributional changes considered in these works are symmetric and following a Normal distribution. In contrast, we pose a flexible distribution that allows for asymmetric fluctuations of idiosyncratic risk that also capture the fact that changes are more likely to be very small or very large compared to a Normal distribution (see evidence in, e.g., [Guvenen *et al.*, 2014](#); [Busch *et al.*, 2022](#)). Importantly, our main goal lies on quantifying the success of the existing tax and transfer system in smoothing the extent and business cycle variation of idiosyncratic risk; different to [Busch and Ludwig \(2024\)](#), who explore the role of remaining higher-order risk in a quantitative model.

Relative to this literature we adopt an analytically tractable modeling framework, which links cyclical idiosyncratic risk to consumption dynamics and welfare. We build on [Heathcote *et al.* \(2014\)](#)’s partial insurance framework. In this sense, we bridge results from the life-cycle literature that focuses on a bundle of self- and family-insurance channels beyond the traditional savings instruments considered in the business-cycle literature ([Blundell *et al.*,](#)

2008; Krueger and Perri, 2006; Kaplan and Violante, 2010a). In contrast to HSV, who pose a process for wages and explicitly model two insurance channels—endogenous labor supply and a progressive tax function—we treat household income as the fundamental source of risk, and incorporate a rich income process with time-varying risk in the spirit of McKay (2017) into the model framework, while retaining analytical tractability.

2 Public Insurance as Partial insurance

Before introducing our full measurement framework, it is useful to briefly discuss established definitions of partial insurance. Theoretical measures and their empirical counterparts involve linking consumption changes to individual income changes. A “natural” reference point is full insurance, which arises in the context of complete markets. Mace (199x) derives estimation equations from the theoretical framework and puts the hypothesis of full insurance to test in individual-level panel data. Another early reference point in the literature is the Permanent-Income-Hypothesis, which implies that given linear marginal utility, under the availability of flexible smoothing devices, individuals smooth out transitory shocks to their income, while permanent shocks translate one-for-one to consumption adjustments.

In the absence of complete markets that allow for full insurance against idiosyncratic risk, individual risk is partially insurance against. Empirical evidence builds around specifying consumption as a function of income shocks, typically transitory and persistent ones. Prominently, Blundell *et al.* (2008) use an estimation equation of the form

$$\Delta \log c_{it} = \beta_0 + \beta_1 \eta_{it} + \beta_2 \varepsilon_{it} + u_{it}, \quad (1)$$

where η and ε are permanent and transitory shocks to individual i ’s income, respectively. If shocks were directly observable in the data, the *pass-through* of a permanent shock to consumption could be pinned down using OLS as

$$\beta_1 = \frac{\text{cov}(\Delta \log c_{it}, \eta_{it})}{\text{var}(\eta_{it})}, \quad \beta_2 = \frac{\text{cov}(\Delta \log c_{it}, \varepsilon_{it})}{\text{var}(\varepsilon_{it})} \quad (2)$$

and likewise for transitory shocks. Given these coefficients, the parameter of partial insurance is given by

$$\lambda_{perm} = 1 - \beta_1 \tag{3}$$

$$\lambda_{trans} = 1 - \beta_2. \tag{4}$$

Given that shocks are not observable, in practice the consumption function is specified together with a stochastic income process. The pass-through coefficients β_1 and β_2 are then identified together with the variances of the shocks from a set of population moments. Accordingly, [Blundell *et al.* \(2008\)](#) establish an identification strategy, and estimate pass-through coefficients for different measures of income in the Panel Study of Income Dynamics. Of particular interest to us are the estimations using household-level gross income and disposable income, respectively.

These measures of pass-through for the two income measures are directly linked in the following sense: consumption is a function of disposable income, and thus consumption reacts to changes in gross income through adjustments of disposable income. In other words, the total pass-through from gross income to consumption combines the pass-through from gross income to disposable income with the pass-through from disposable income to consumption. To structure this, think of a tax function a la [Bénabou \(2000\)](#) and [Bénabou \(2002\)](#), as more recently used by, e.g., [Heathcote *et al.* \(2017\)](#), where net tax revenues at income level y are given by $T(y) = y - \phi y^{1-\tau}$, so that disposable income is given by:

$$y^{disp} = \phi y^{1-\tau}. \tag{5}$$

The progressivity parameter, τ ,² directly translates into the elasticity of disposable income with respect to gross income, $1 - \tau$: $\Delta \log y^{disp} = (1 - \tau)\Delta \log y$, and thus

$$\frac{\text{cov}(\Delta \log c, \Delta \log y^{disp})}{\text{var}(\Delta \log y^{disp})} = \frac{(1 - \tau)\text{cov}(\Delta \log c, \Delta \log y)}{(1 - \tau)^2 \text{var}(\Delta \log y)}, \quad (6)$$

$$\Longleftrightarrow \lambda = 1 - (1 - \tau)(1 - \lambda^{disp}). \quad (7)$$

It is useful to consider some bounds for common reference values:

$$\lambda = \begin{cases} \tau & \text{if } \lambda^{disp} = 0 \text{ (no self-insurance)} \\ 1 & \text{if } \lambda^{disp} = 1 \text{ (full self-insurance)} \end{cases} \quad (8)$$

where λ^{disp} is the degree of partial consumption insurance against shocks to disposable income.

In other words, if agents are able to fully self-insure, then the degree of public insurance is irrelevant. If agents have no ability to self-insure (they are hand-to-mouth), then total insurance is equal to public insurance, which is exactly equal to the degree of progressivity.³ Previous studies show that most agents are somewhere in between (e.g., [Blundell et al., 2008](#), estimate a $\lambda^{disp} = 0.64$ using panel data for the United States from the PSID).

Estimation of the pass-through coefficients uses panel data on all three measures: gross income, disposable income, and consumption. While administrative sources of income, before and after taxes and transfers, have become widely available in recent years, data on consumption is still scarce and subject to measurement issues.⁴ The sketched tax function on the other

²If $\tau > 0$, marginal tax rates exceed average rates and hence the tax and transfer system is considered progressive. Conversely, when $\tau = 0$ the tax and transfer scheme is flat.

³Of course, generally the degree of self-insurance is endogenous with respect to existing policies that determine the degree of public insurance.

⁴In survey data, measurement error and low frequency pose challenges. In administrative data, imputations are required. In bank records, samples are rarely representative. And, in all cases, a pervasive measurement issue regardless of the source is the disconnect between expenditures and consumption, particularly serious for durable consumption.

hand works without consumption data by simply linking pre- and post-government incomes. This achievement, however, is bought by strong parametric assumptions regarding exactly this link.

The tax function implies a mechanical link between the distributions of changes in gross income and disposable income. In particular, the variance of changes is scaled by $(1 - \tau)^2$, see (6). The skewness of changes is the same. In contrast, in our analysis, we flexibly capture distributional features of gross income and disposable income changes without posing any such restriction. We now move to establishing a tool that allows us to assign an insurance value to the tax and transfer system without restricting the relationship between pre- and post-government income and without resorting to consumption data.

3 Quantitative Model: A Measurement Device

3.1 Model Economy

Endowment structure and preferences. We consider a stochastic endowment economy, which is populated by a continuum of islands, each of which is in turn populated by a continuum of agents. There are two types of shocks: one common to all members of an island and the other purely idiosyncratic. The within-island shocks wash out on the island, the island-level shocks wash out across islands, such that there is no aggregate risk to total endowment. An island refers to a group of agents that are described by the same history of island-level shocks (common to all members of the group).

Islands can be thought of as a network of family members, who perfectly share the risks faced by each individual. If, for example, all family members work in the same industry and live in the same region, there will be shocks that hit every member equally and hence cannot be insured within the family network. Importantly for the quantitative analysis, there is no need to define empirical counterparts to the model islands.

Specifically, individual income (endowments) is assumed to follow

$$\begin{aligned}
y_t &= y_t^{island} + y_t^{idio} \\
y_t^i &= z_t^i + \varepsilon_t^i, \quad \varepsilon_t^i \sim F_{\varepsilon,t}^i, \quad \text{for } i \in \{island, idio\} \\
z_t^i &= z_{t-1}^i + \eta_t^i, \quad \eta_t^i \sim F_{\eta,t}^i, \quad \text{for } i \in \{island, idio\}
\end{aligned} \tag{9}$$

where z_t^i and ε_t^i for $i \in \{island, idio\}$ denote the island-level and idiosyncratic permanent and transitory components of income. All stochastic components of income are independent and normalized such that $\int \exp(x_t^i) dF_{x,t}^i = 1$ for $i \in \{island, idio\}$ and $x \in \{\varepsilon, \eta\}$.

Agents live finite lives. Each period a mass $(1 - \delta)$ of newborns enters the economy with age 0. The probability of survival from age a to age $a + 1$ is constant at δ . Agents maximize discounted lifetime utility, whereby we assume time- and state-separable preferences. For the per period utility function, we use log utility as the benchmark: $U(c_t) = \log(c_t)$. We also study the importance of this assumption and inspect the role of stronger risk attitudes by using an alternative specification with a CRRA per period utility function with parameter of relative risk aversion larger than 1. The discount factor β is constant across the population.

Age 0 agents entering in year τ hold zero financial wealth and are allocated to an island of agents which then share the same sequence of island-level shocks $\{\eta_t^{island}, \varepsilon_t^{island}\}_{t=\tau}^{\infty}$.

Asset markets and equilibrium. Every period agents engage in asset trade. There is a full set of state-contingent claims available to agents within islands. Claims are in zero net supply. Across islands, agents cannot trade claims contingent on the island-level shocks. This restriction on available assets implies that in equilibrium island-level shocks remain uninsured, while within-island shocks are fully insured, and risk is shared by all individuals on an island. In other words, a no-trade equilibrium in the spirit of [Constantinides and Duffie \(1996\)](#) exists. While in their model, idiosyncratic endowment shocks remain fully uninsured in this no-trade equilibrium, the equilibrium in our model entails partial insurance: there is no asset trade across islands, while agents within an island insure themselves perfectly against the individual-specific shocks. This mimics the result in [Heathcote *et al.* \(2014\)](#).

In equilibrium, log consumption and consumption change are given by⁵

$$\log c_t(\mathbf{x}_t, y_t^{idio}) = y_t^{island} + \log \int \exp(y_t^{idio}) dF_{y^{idio},t}^a \quad (10)$$

$$\begin{aligned} \Delta \log c_t &= \Delta y_t^{island} + \log \frac{\int \exp(\eta_t^{idio}) dF_{\eta,t}^{idio} \int \exp(\varepsilon_t^{idio}) dF_{\varepsilon,t}^{idio}}{\int \exp(\varepsilon_{t-1}^{idio}) dF_{\varepsilon,t-1}^{idio}} \\ &= \eta_t^{island} + \Delta \varepsilon_t^{island} + \log \frac{\int \exp(\eta_t^{idio}) dF_{\eta,t}^{idio} \int \exp(\varepsilon_t^{idio}) dF_{\varepsilon,t}^{idio}}{\int \exp(\varepsilon_{t-1}^{idio}) dF_{\varepsilon,t-1}^{idio}} \\ &= \eta_t^{island} + \Delta \varepsilon_t^{island}, \end{aligned} \quad (11)$$

where $\Delta \log c_t = \log c_t - \log c_{t-1}$ and $\Delta \varepsilon_t^{island} = \varepsilon_t^{island} - \varepsilon_{t-1}^{island}$. The above equation summarizes the major advantage—relative to standard incomplete market models—of introducing the partial insurance framework by the abstraction of islands: it allows for an analytical solution in which consumption changes are expressible as an explicit function of idiosyncratic shocks. That is, given an endowment process, we can directly calculate the consumption level and change implied by the model.

Note that the uninsurable, island-level shocks translates one-for-one to consumption. The individual realizations of the two insurable shocks, however, do not affect consumption: given perfect risk-sharing, all members of an island consume the mean realization of these shocks.

Degree of partial insurance. [introduced above, refer there] [Blundell et al. \(2008\)](#) introduce a *pass-through coefficient* for a given shock to denote the fraction of the shock that translates into consumption changes. We use the model equivalent of the pass-through to define the insurance coefficients against transitory and permanent shocks, where in the same spirit as, e.g., [Kaplan and Violante \(2010b\)](#) we assume that shock components are observed, i.e., transitory and permanent shocks can be told apart by model agents. As is clear from (11), in our model framework, *island*-shocks translate one-for-one to consumption—the pass-through of shock to consumption is one—and *idio*-shocks do not translate into consumption—the pass-through of shock to consumption is zero. We thus consider the pass-through not to one of those shocks, but to the combined *island* and *idio*-shocks. First, consider the pass-through

⁵The derivation of consumption outlined in [Heathcote et al. \(2014\)](#) carries over one-for-one to our model version, simplified by the fact that we do not have a tax function nor endogenous labor supply.

of overall income changes, i.e., Δy_t :

$$1 - \lambda = \frac{\text{cov}(\Delta \log c_t, \Delta y_t)}{\text{var}(\Delta y_t)} = \frac{\text{cov}(\Delta y_t^{island}, \Delta y_t)}{\text{var}(\Delta y_t)} = \frac{\text{var}(\Delta y_t^{island})}{\text{var}(\Delta y_t)}. \quad (12)$$

Thus, the ratio of the variance of *island* shocks to total shocks determines the pass-through of income changes to consumption—and conversely, the ratio of the variance of *idio* shocks to total shocks gives the degree of partial insurance, λ .

We then split the above up, explicitly considering different degrees of insurance against transitory and permanent shocks, which gives

$$\begin{aligned} 1 - \lambda_{trans} &= \frac{\text{cov}(\Delta \log c_t, \varepsilon_t)}{\text{var}(\varepsilon_t)} \\ &= \frac{\text{cov}(\Delta \varepsilon_t^{island}, \varepsilon_t)}{\text{var}(\varepsilon_t)} = \frac{\text{cov}(\varepsilon_t^{island} - \varepsilon_{t-1}^{island}, \varepsilon_t^{island} + \varepsilon_t^{idio})}{\text{var}(\varepsilon_t)} \\ &= \frac{\text{var}(\varepsilon_t^{island})}{\text{var}(\varepsilon_t^{island} + \varepsilon_t^{idio})} = \frac{\text{var}(\varepsilon_t^{island})}{\text{var}(\varepsilon_t^{island}) + \text{var}(\varepsilon_t^{idio})} \end{aligned} \quad (13)$$

and

$$\begin{aligned} 1 - \lambda_{perm} &= \frac{\text{cov}(\Delta \log c_t, \eta_t)}{\text{var}(\eta_t)} \\ &= \frac{\text{cov}(\eta_t^{island}, \eta_t)}{\text{var}(\eta_t)} = \frac{\text{cov}(\eta_t^{island}, \eta_t^{island} + \eta_t^{idio})}{\text{var}(\eta_t)} \\ &= \frac{\text{var}(\eta_t^{island})}{\text{var}(\eta_t^{island} + \eta_t^{idio})} = \frac{\text{var}(\eta_t^{island})}{\text{var}(\eta_t^{island}) + \text{var}(\eta_t^{idio})}, \end{aligned} \quad (14)$$

such that the degree of partial insurance against permanent shocks, λ_{perm} is given by the fraction of the variance of permanent shocks attributable to the *idio*-component, and the degree of partial insurance against transitory shocks is given by the fraction of the variance of transitory shocks attributable to the *idio*-component.

Tax and transfer system. We then introduce a tax and transfer system that alters the endowment stream faced by agents. We do not explicitly model the tax system, but retain full flexibility about its nature—i.e., we do not make any functional form assumption. Instead, we consider a second scenario in which agents face income stream (9), but with different

distributions of shocks. Importantly, we maintain the normalization that $\int \exp(x_t^i) dF_{x,t}^i = 1$ for $i \in \{\text{island}, \text{idio}\}$ and $x \in \{\varepsilon, \eta\}$, which means that we consider a tax and transfer system that cross-sectionally redistributes endowments, which means that, e.g., we do not allow for wasteful government consumption nor for transfer payments financed by debt.

3.2 Insurance Value of Tax and Transfer System

We now use the model structure outlined above in order to back out the degree of partial insurance provided by the tax and transfer system. To this end, we consider the following experiment. Agents live in one of two possible scenarios that differ in the endowment streams that agents face. In the first, the endowment stream describes pre-government incomes. In the second, the endowment stream describes post-government incomes. We then assume a degree of partial insurance against (total) individual shocks in the post-government scenario—i.e., we assume values for λ_{trans}^{post} and λ_{perm}^{post} . Given this assumed amount of partial insurance, we obtain stochastic consumption streams per equation (11).

We then find the degree of partial insurance in the pre-government scenario that makes agents ex ante indifferent to living in the post-government scenario (for the given degree of insurance in the latter). Given that there are two types of shocks, in principle multiple combinations of $\{\lambda_{trans}^{pre}, \lambda_{perm}^{pre}\}$ can exist that make agents indifferent. We assume that $\lambda_{trans}^{pre} = \lambda_{trans}^{post} = 1$. Thus, we assume that transitory shocks are well insured and do not pass-through to consumption. In the abstraction of the island model this shows by having no island-level shocks, and instead all transitory shocks happen purely within islands, and thus can be insured away fully by agents trading state-contingent claims. As mentioned in the introduction, note that incomplete market models typically find very high insurance against transitory shocks through private savings alone (Busch and Ludwig, 2024; De Nardi *et al.*, 2020; Kaplan and Violante, 2010b), thus full insurance appears to be a plausible assumption.

This leaves partial insurance against permanent risk as the relevant margin of the model. Given an assumption regarding λ_{perm}^{post} , we find the λ_{perm}^{pre} that makes agents indifferent. In our benchmark calculation, we assume that $\lambda_{perm}^{post} = 0$. This assumption is motivated by empirical results in Blundell *et al.* (2016), who find that the degree of partial insurance on top of government and family transfers is very close to zero. The obtained λ^{pre} can

then be interpreted as the degree of partial insurance provided by the government under the assumption that there is no additional partial insurance. We show below that the assumption on λ_{perm}^{post} is not strong.

4 Pre- and Post-Government Income in Sweden

Given the measurement device provided by the model outlined above, we are set for evaluating the degree of partial insurance provided by the tax and transfer system. The two empirical ingredients necessary are two stochastic income streams: one that captures the regularities of pre-government income, and one that captures the regularities of post-government income. We estimate these using Swedish data moments.

4.1 Estimated Income Processes

Let y_t^{pre} and y_t^{post} denote log of pre- and post-government household income, respectively. We assume that it follows the following permanent-transitory process (where we drop the explicit reference to pre or post-government income):

$$\begin{aligned} y_t &= z_t + \varepsilon_t \\ z_t &= z_{t-1} + \eta_t \end{aligned} \tag{15}$$

where ε_t is an *iid* transitory shock, and η_t denotes a permanent shock with time-varying and business-cycle-dependent distribution, modeled as in [McKay \(2017\)](#). We specify the distribution functions such that the process can match excess kurtosis and skewness found in the data.

In particular, the transitory component ε_t is drawn from a mixture of two normals:

$$\varepsilon_t \sim \begin{cases} \mathcal{N}(\bar{\mu}_\varepsilon, \sigma_{\varepsilon,1}^2) & \text{with prob. } p_{\varepsilon,1} \\ \mathcal{N}(\bar{\mu}_\varepsilon, \sigma_{\varepsilon,2}^2) & \text{with prob. } 1 - p_{\varepsilon,1} \end{cases} \tag{16}$$

where $p_{\varepsilon,1}$ denotes the probability of drawing from component 1; $\bar{\mu}_\varepsilon$ is chosen such that $\mathbb{E}[\exp(\varepsilon)] = 1$. The permanent component η_t follows a mixture of three normals:

$$\eta_t \sim \begin{cases} \mathcal{N}(\bar{\mu}_{\eta,t} + \mu_{\eta,1} + \phi_1 x_t, \sigma_{\eta,1}^2) & \text{with prob. } p_{\eta,1} \\ \mathcal{N}(\bar{\mu}_{\eta,t} + \mu_{\eta,2} + \phi_2 x_t, \sigma_{\eta,2}^2) & \text{with prob. } p_{\eta,2} \\ \mathcal{N}(\bar{\mu}_{\eta,t} + \mu_{\eta,3} + \phi_3 x_t, \sigma_{\eta,3}^2) & \text{with prob. } p_{\eta,3} \end{cases} \quad (17)$$

where $p_{\eta,i}$, $i = 1, 2, 3$, denotes the probability of drawing from component i , where $\sum_{i=1}^3 p_{\eta,i} = 1$. The parameters ϕ_i determine how strongly aggregate risk as captured by x_t translates into changes of the distribution of idiosyncratic earnings risk. x_t is standardized log GDP growth. As part of our goal is to capture the business-cycle fluctuations of idiosyncratic income risk, we choose $\bar{\mu}_{\eta,t}$ such that $\mathbb{E}[\exp(\eta_t)] = 1$. In the estimation, we then shift the distribution so as to impose the mean of medium-run (3-year) income changes to be as in the data. We use GDP growth as the empirical measure of aggregate fluctuations in order to make the quantitative results easily interpretable. Over the period of estimation, the average GDP growth rate is 2.15% with a standard deviation of about 2.35%.

Estimation of process. We estimate the set of parameters $\chi = \{\chi_{trans}, \chi_{perm}\}$ where

$$\chi_{trans} = \{\sigma_{\varepsilon,1}, \sigma_{\varepsilon,2}, p_{\varepsilon,1}\} \quad (18)$$

$$\chi_{perm} = \{\mu_{\eta,2}, \mu_{\eta,3}, \sigma_{\eta,1}, \sigma_{\eta,2}, p_{\eta,1}, p_{\eta,2}, \phi_2, \phi_3\} \quad (19)$$

by the simulated method of moments (SMM).⁶ We target the time series of L9050 and L5010⁷ of the 1, 3, and 5-year earnings changes distribution, the average of the Crow-Siddiqui measure of kurtosis of 1-, 3-, and 5-year changes, as well as the age profile of the cross-sectional

⁶For identification purposes, we impose $\mu_{\eta,2} \geq 0$, $\mu_{\eta,3} \leq 0$, and $\phi_1 = 0$. With this assumption, the time-varying means of the three mixtures will control the center, right tail, and left tail of the distribution of η , respectively. For practical purposes, we further assume $p_{\eta,2} = p_{\eta,3}$, $\sigma_{\eta,2} = \sigma_{\eta,3}$.

⁷L9050 = $P_{90} - P_{50}$ denotes the difference between the 90th and 50th percentiles, and likewise L5010 = $P_{50} - P_{10}$.

variance from ages 25 to 60. The Crow-Siddiqui measure of kurtosis ([Crow and Siddiqui, 1967](#)) is defined as $\mathcal{CS} = \frac{(P97.5 - P2.5)}{(P75 - P25)}$. This gives 213 moments for our estimation of the income process for Sweden.

To construct the simulated income profiles over time, we write earnings growth as a function of the shocks, using equation (15):

$$y_t - y_{t-s} = \varepsilon_t - \varepsilon_{t-s} + \sum_{j=0}^{s-1} \eta_{t-j}, \quad (20)$$

for different horizons $s = 1, 3, 5$. The simulated series of the life-cycle variance profile of log earnings is computed as follows. We assume a time-invariant distribution of shocks by imposing $x_t = 0 \forall t$. Notice that this assumes that the variance accumulates linearly over the life cycle. We then normalize the series so that the variance at age 25 in the simulation is 0. Finally, we rescale the resulting simulated profile to exhibit the same mean as its empirical counterpart.

We simulate these profiles $R = 10$ times for $I = 100,000$ individuals and compute the moments corresponding to the aforementioned targets. To find $\hat{\chi}$, we minimize the average scaled distance between the simulated and empirical moments. A weighting matrix is used to scale the life-cycle profile. In particular, we weight the variance profile with 20% and the remaining moments with 80%. For the optimization part, we use a global version of the Nelder-Mead algorithm with several quasi-random restarts, as described in [Guvenen \(2011\)](#).

Let c_n^m denote the empirical moment n ($n = 1, \dots, N$) that corresponds to cross-sectional target $m \in \{L5010(\Delta^1 y_t), L5010(\Delta^3 y_t), L5010(\Delta^5 y_t), \dots, var(y_{age=25}), \dots, var(y_{age=60})\}$. In each simulation, we draw a matrix of random variables $X_r = \{\varepsilon_1^i, \varepsilon_2^i, \dots, \varepsilon_T^i, \eta_1^i, \dots, \eta_T^i\}_{i=1}^I$ where T denotes the last year available in the data. For each simulation, we calculate the respective simulated moments $d_n^m(\chi, X_r)$ given the parameter vector χ .

We minimize the scaled deviation $F(\chi)$ between each data and simulated moment

$$\min_{\chi} F(\chi)' W F(\chi)$$

where F is defined as

$$F_n(\chi) = \frac{d_n^m(\chi) - c_n^m}{|c_n^m|}$$

$$d_n^m(\chi) = \frac{1}{R} \sum_{r=1}^R d_n^m(\chi, X_r)$$

Parameter estimates. Table 1 shows the parameter estimates. To illustrate the magnitude of the estimated swings in the distribution of idiosyncratic risk, consider the time period around the Great Recession. During those years, the GDP growth rate plummets to a negative GDP growth of -5.04% in 2008 (about three standard deviations below the average), recovers to a strong 6.59% in 2009 (about 2 standard deviations above the average), followed by an about average growth year in 2010 with 2.49% . Over the course of these three years, the distribution of individual earnings changes is estimated to vary markedly as shown in Figure 1, which plots the distribution of the permanent component of income changes, η_t for both pre-government and post-government income. Each panel shows the simulated distribution for the estimated mixture of Normals in red (histogram), as well as the density of the corresponding Normal distribution with the same mean and variance (but with a skewness of 0 and a kurtosis of 3). In the plots, we use the normalization such that $E[\exp(\eta_t)] = 1$. For completeness, Figures 3 and 4 in Appendix A show the simulated moments at these parameters together with the empirical moments.

Table 1: Estimated Parameter Values

Parameter	Description	Pre-Gov.	Post-Gov.
$p_{\varepsilon,1}$	Mixture prob. of ε distribution	0.892	0.877
$\sigma_{\varepsilon,1}$	Std. dev. of ε distribution mix. comp. 1	0.055	0.047
$\sigma_{\varepsilon,2}$	Std. dev. of ε distribution mix. comp. 2	0.628	0.401
$p_{\eta,1}$	Mixture prob. of η distribution mix. comp. 1	0.981	0.981
$p_{\eta,2}$	Mixture prob. of η distribution mix. comp. 2	0.010	0.009
$p_{\eta,3}$	Mixture prob. of η distribution mix. comp. 3	0.010	0.009
$\sigma_{\eta,1}$	Std. dev. of η distribution mix. comp. 1	0.086	0.057
$\sigma_{\eta,2}$	Std. dev. of η distribution mix. comp. 2	0.020	0.009
$\sigma_{\eta,3}$	Std. dev. of η distribution mix. comp. 3	0.020	0.009
$\mu_{\eta,2}$	Mean of mixt. comp. 2 of η distribution	0.002	0.008
$\mu_{\eta,3}$	Mean of mixt. comp. 3 of η distribution	-0.158	-0.065
ϕ_2	Aggregate risk transmission mixt. comp. 2	1.186	1.240
ϕ_3	Aggregate risk transmission mixt. comp. 3	0.467	0.229
M	# moments targeted in estimation	213	213

Note: Estimated parameters for gross household labor income (Pre-Gov.) and household income after taxes and transfers (Post-Gov.) in Sweden.

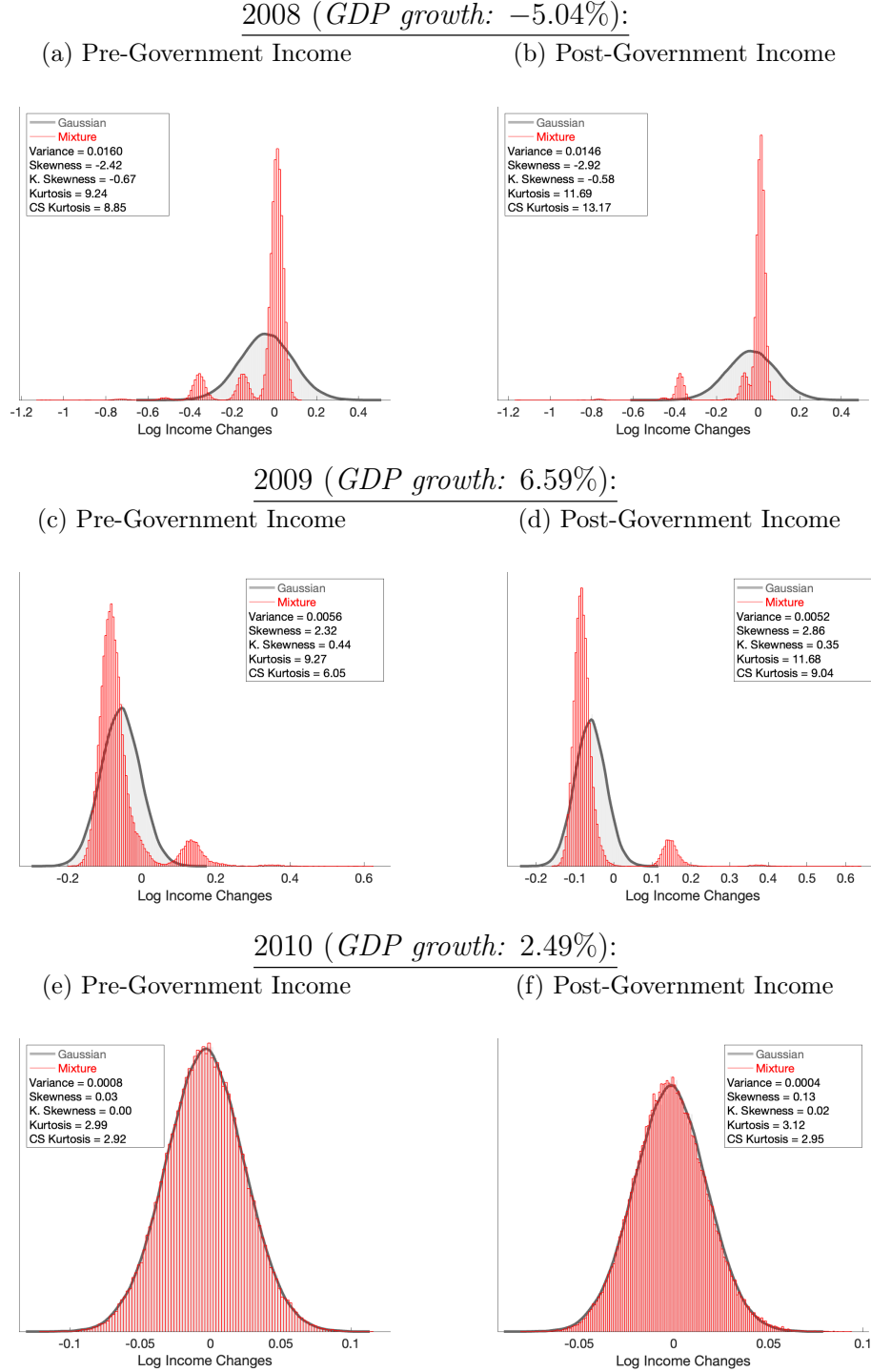
As captured in the three rows of the figure, the distribution of permanent income changes varies over the cycle in an asymmetric way for both measures of income (pre- and post-government). Strong negative GDP growth (as in 2008) goes hand-in-hand with a left-skewed distribution as captured by a negative coefficient of skewness, while strong positive GDP growth (as in 2009) comes with a right-skewed distribution. In about average growth times, the idiosyncratic distribution turns out to be well captured by a Gaussian distribution—and while it is already very narrow for pre-government income, the tax and transfer system compresses the distribution even more: the variance is halved.

In an expansionary year (like 2009) the distribution turns into a right-skewed distribution, as captured by a positive coefficient of skewness (the third standardized moment), and also a positive measure of Kelley’s skewness, which is a percentile-based measure of skewness calculated as $\mathcal{KS} = \frac{(P90-P50)-(P50-P10)}{P90-P10}$. $\mathcal{KS}$ takes on values $\in (-1, 1)$, and captures the relative size of the left and right tails in overall dispersion. Kelley’s skewness is a useful statistic to interpret the magnitude of the change in the distribution over the cycle. For pre-

government income, the value of $\mathcal{KS} = 0.44$ for 2009 indicates that $(P90 - P50)$ accounts for 72% of the $(P90 - P10)$ dispersion.⁸ On the other hand, in 2008 the value of $\mathcal{KS} = -0.67$ indicates that $(P90 - P50)$ accounts for only 16.5% of the $(P90 - P10)$ dispersion.

⁸Note that $P90 - P50 = 0.5 + \frac{\mathcal{KS}}{2}$.

Figure 1: Cross-Sectional Distribution of Permanent Income Changes



Note: Each figure shows the distribution of simulated (pre-government) permanent income changes η in red (histogram), as well as the density of the corresponding normal distribution with same mean and variance but with 0 Skewness and a Kurtosis of 3. The different figures correspond to three years that are representative of different states of the business cycle.

The tax and transfer system dampens the swings of skewness over the business cycle, which is captured in the parameter estimates for ϕ_2 and ϕ_3 in Table 1. This is reflected in the distributions plotted for years 2008 and 2009. Also for post-government income, $\mathcal{KS}$ changes from negative in 2008 to positive in 2009. However, the difference is less pronounced than for pre-government income. In 2008, $\mathcal{KS} = -0.58$ indicates that $(P90 - P50)$ accounts for 21% of the $(P90 - P10)$ dispersion. In 2009, $\mathcal{KS} = 0.35$ indicates that $(P90 - P50)$ accounts for 67.5%. Furthermore, the distribution is leptokurtic for both income measures in 2008 and 2009, with a somewhat higher kurtosis for post-government income, which implies that the tax and transfer system overall increases the concentration of the distribution.

To sum up, taxes and transfers, (i.), reduce overall dispersion of income changes, (ii.), reduce the cyclicity of dispersion and skewness, (iii.), increase concentration of income changes in both contractionary and expansionary years. The question we turn to now is: how do households value this?

4.2 Cross-Sectional Distribution Over the Life Cycle

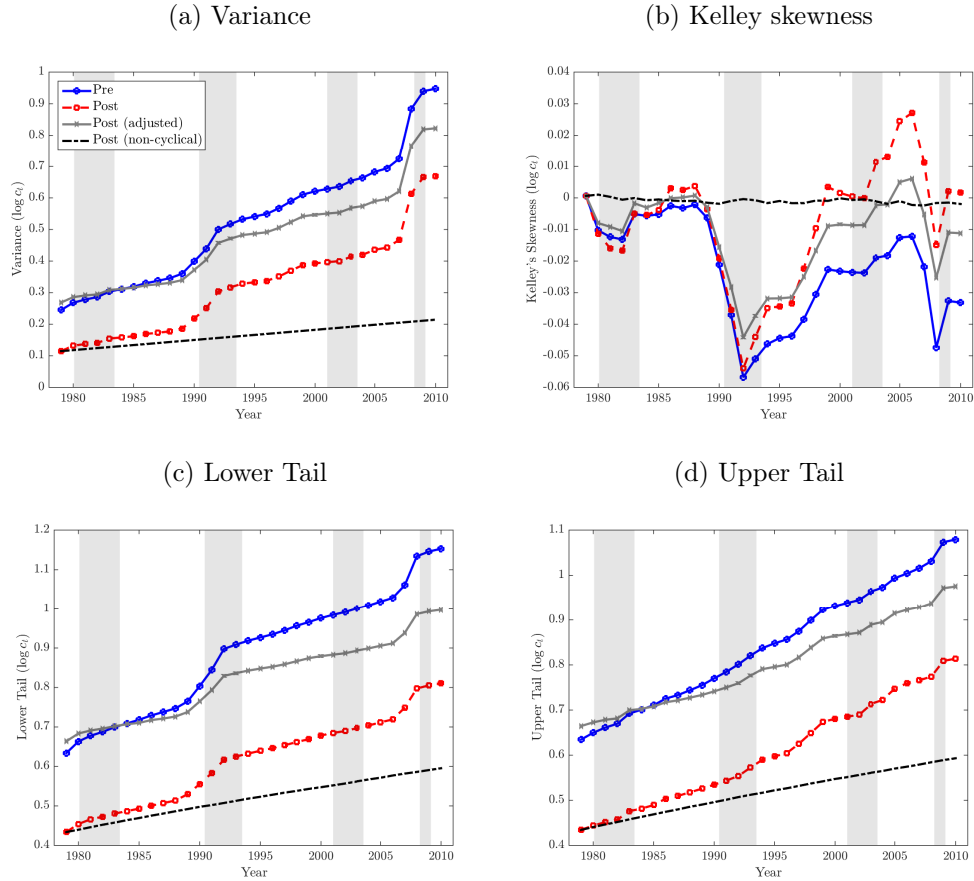
From an ex-ante perspective, the distribution of possible income streams that can realize over the life cycle are relevant when it comes to the assessment of different risk scenarios. Given our assumption on full insurance against transitory shocks, the permanent shocks are relevant for the consumption distribution, and thus for welfare. Those shocks accumulate and generate a distribution that widens as a cohort ages.

We now consider a cohort of agents that lives through the Swedish macroeconomic history captured by the process of x_t that is used in the estimation of the income process. This way, we obtain a distribution of possible paths of the permanent income component for individuals entering the Swedish economy in year 1978, who subsequently receive stochastic income according to the estimated income process. For a given degree of partial insurance, we can then generate the corresponding consumption paths by simulating forward using equation (11).

Consider the blue line in panel (a) in Figure 2: it shows how the variance of (the model-constructed) cross-sectional permanent income component of pre-government income evolves for the cohort living through the Swedish macroeconomic history. During the contractions of

the early 1990s and the late 2000s, the distribution of shocks becomes more dispersed, and thus the increase of the cross-sectional variance gets steeper. Panels (c) and (d) show that this increase in contractions happens stronger in the lower tail, which reflects an asymmetric swing of the distribution, that also manifests itself in the evolution of cross-sectional skewness, which is shown in panel (b): it tends to get more negative in contractions, and more positive in expansions.

Figure 2: Cross-Sectional Distribution of Permanent Income



Note: Each figure shows a moment of the simulated cross-sectional distribution of permanent income for a cohort that lives through the Swedish macroeconomic history and faces, (i), the estimated pre-government income process; (ii), the estimated post-government income process; (iii), the post-government income process adjusted for initial variance; or, (iv), the post-government income process that eliminates all cyclicity of the distribution of shocks.

In each of the four panels of Figure 2, the red line reports the cross-sectional moments of the permanent component of post-government income for the same cohort. In line with the discussion of the estimated permanent income change component in the previous section, the first key difference is that the overall dispersion at every age is smaller (see panel a). Second, in the years leading up to the recession of the early 1990s, the asymmetry as measured by Kelley’s skewness behaves very similarly; in the subsequent recovery Kelley’s skewness of post-government income gets less and less negative and turns positive around the mid-2000s.

5 Measured Insurance Value of Taxes and Transfers

5.1 Overall Level of Insurance

We now employ the model measure derived in Section 3.2 to derive the degree of partial insurance against permanent income risk, λ_{perm}^{pre} , implied by the tax and transfer system. Thus, in line with the description in Section 3.2, we the goal is to find the λ_{perm}^{pre} , which yields a consumption stream that makes households indifferent to facing the post-government income stream—with a given amount of partial insurance when facing the latter.

For a given λ_{perm}^{pre} , we scale the estimated parameters of the permanent shocks such that the variance of the resulting distribution for η_t^{idio} is equal to fraction λ_{perm}^{post} of the overall variance of the permanent shock η . The scaling is such that the shape of the distribution as captured by the coefficient of skewness remains the same. We normalize such that $E[\exp(\eta^{island})] = E[\exp(\eta^{idio})] = 1$. Under log utility we find $\lambda_{perm}^{pre} = 0.43$, which means that the existing tax and transfer schedule in Sweden corresponds to insuring households against 43% of permanent shocks to household labor income, as shown in Table 2.

In order to assess the magnitude of this degree of partial insurance in terms of welfare, we use the model to calculate the consumption equivalent variation (CEV) that makes agents in the scenario with the *pre-government income stream and no partial insurance* indifferent to the world with the *pre-government income stream and partial insurance of the size given by λ_{perm}^{pre}* . The 43% partial insurance translates into a CEV of 14.3% when assuming log utility. Hence, the existing tax and transfer system provides sizable insurance. Note that

this calculation abstracts from any first-order effects: both a potential level effect of the tax and transfer system on the aggregate income of a given cohort and the cyclical variation in average income changes are taken out of the equation.

5.2 Decomposition of Insurance Channels

Initial dispersion. When interpreting these results, it is important to notice that government policy reduces the overall level of cross-sectional dispersion, and the cyclical variation of shocks. In order to differentiate those two smoothing effects, we impose in a second run of the same experiment that the cross-sectional variance at age 25 (when agents are born in the model) is the same as for the pre-government process. The moments of the resulting permanent income process are shown as the gray lines in Figure 2. We now obtain $\lambda_{perm}^{pre} = 0.06$, i.e., moving from the pre- to the post-government income stream adjusted to the same initial variance amounts to partial insurance of 6%, which translates into a CEV of about 1.3%.

Gain of eliminating cyclical risk. Given the already sizable insurance, what is the scope of additional government policy as a means of insurance against cyclical risk? In order to approach this question, we consider the same experiment for a counterfactual income process. Assume that on top of what the government already does, cyclical risk is completely shut down for the post-government income stream. For this experiment, we set $\phi_2 = \phi_3 = 0$, thus imposing the distribution of idiosyncratic income changes that corresponds to periods of average GDP growth. This yields the profiles of cross-sectional moments shown by the dashed lines in Figure 2. This implies an even stronger degree of insurance of about 64% (or 27% when adjusting for initial variance at age 25). Considering the CEV connected to those insurance parameters, the scope of additional insurance is sizable: through the lens of the model, when adjusting for initial variance effects, an additional welfare gain of about 4.6 percentage points is possible.

Role of higher-order moments. In the estimation of the income process we were careful to match not only the dispersion of income changes, but also measures of skewness and kurtosis, i.e., higher-order moments of the distributions of individual income changes over

Table 2: Partial Insurance and Welfare Gains of the Tax and Transfer System

Scenario	λ_{perm}^{pre}	CEV	λ_{perm}^{pre} (cycl.)	CEV (cycl.)
	log utility			
Pre to Post	43%	14.26%	6%	1.28%
<i>Gaussian</i>	43%	15.52%	7%	2.97%
Pre to Post*	64%	17.53%	27%	5.91%
	65%	20.60%	29%	11.15%
	<i>CRRA w/ Risk Aversion = 2</i>			
Pre to Post	36%	32.65%	5%	3.03%
	42%	34.80%	8%	7.23%
Pre to Post*	66%	46.34%	34%	19.13%
	65%	47.57%	30%	25.80%

Note: The term λ_{perm}^{pre} denotes the degree of partial insurance against permanent shocks. * indicates that the cyclical of the permanent shocks is shut down. See text for details on the scenarios. The CEV columns denote the corresponding consumption equivalent variation associated with the change from the world with the pre-government income stream and no partial insurance to a world with the pre-government income stream and partial insurance of the size given by λ_{perm}^{pre} .

the business cycle. As discussed in Section 4, those moments capture salient features of how the distribution varies over the business cycle, as it becomes more left-skewed in contractions. Thus, the next question we ask is whether for our model measure of partial insurance it is relevant to take those higher-order moments (and their cyclical changes) into account or not.

Thus, we now reconsider the exercise, but now assume that agents are exposed to Gaussian earnings processes that share the first and second-moment properties with the estimated pre- and post-government income processes, respectively, but have zero skewness and a kurtosis of 3. Notably, the variance still co-moves with the aggregate state of the economy, as it does in the benchmark case. Note that this implies that the dispersion evolves as displayed in panel (a) of Figure 2, but Kelley's skewness is zero throughout.

The gray rows in Table 2 show the results that correspond to the exact same exercises as in the benchmark analysis, but for the Gaussian shock distributions. There are two take-aways. First, the measured insurance values (and their reflections in CEVs) are of roughly the same magnitude for the overall insurance value of taxes and transfers. Second, the insurance gain against cyclical risk translates into about twice under a Gaussian distribution (2.97% vs. 1.28%). Thus, not taking into account skewness and kurtosis of the distribution of idiosyncratic risk, one would overestimate the insurance value of the existing tax and transfer system. Likewise, the potential additional gain of completely eliminating cyclical variation of idiosyncratic risk is about twice as high: a total gain of 11.15% vs. a total gain of 5.91%.

Role of risk attitudes. So far, we made the assumption that agents have log utility (relative risk aversion of 1). Preferences that feature a constant relative risk aversion larger than 1 are widely used in macroeconomics, and in incomplete market models in particular. The bottom half of Table 2 reports the results for the case of a parameter of relative risk aversion of 2, a standard value. In the context of the analysis it is important to note that this parameter pins down relative risk attitudes also against higher-order risk, which are relevant in order for skewness and kurtosis of the distribution to matter for utility (see detailed discussions in, e.g., [Eeckhoudt, 2012](#); [Busch and Ludwig, 2024](#)). Three patterns emerge. First, the insurance value of the tax and transfer system against total earnings is in general smaller than under risk aversion of 1. Second, however, the CEV of insuring income risk is larger. Third, when focusing on the cyclical component of earnings shocks, both the insurance and welfare gains from taxes and transfers are larger than in the benchmark counterpart. The importance of taking into account higher-order moments (vs. a Gaussian distribution) holds for the stronger risk attitudes.

Role of full pass-through of post-government income. In our benchmark analysis, we derive the consumption profile for households facing the post-government income stream under the assumption of no further partial insurance, i.e., $\lambda_{perm}^{post} = 0$. Given this assumption, we then derive the degree of partial insurance that delivers a consumption stream that makes households indifferent when they face the pre-government income stream. We now explore robustness of the approach with respect to this assumption. For this, we assume that instead,

10% of permanent shocks to post-government income are insured. This delivers a slightly somewhat less dispersed consumption profile. We then evaluate the degree of partial insurance against pre-government income that makes households indifferent; and also repeat the same additional calculations we did for the benchmark case. Results are reported in Table 3.

Table 3: Partial Insurance and Welfare Gains of the Tax and Transfer System ($\lambda_{perm}^{pos} = 0.1$)

Scenario	λ^{gov}	λ_{perm}^{pre}	CEV	$\lambda^{gov}(\text{cycl.})$	$\lambda_{perm}^{pre}(\text{cycl.})$	$CEV(\text{cycl.})$
log utility						
Pre to Post	43%	49%	15.13%	7%	16%	3.29%
<i>Gaussian</i>	43%	49%	16.33%	7%	16%	4.88%
Pre to Post*	64%	68%	18.09%	28%	35%	7.53%
	64%	68%	20.92%	29%	36%	12.36%
<i>CRRA w/ Risk Aversion = 2</i>						
Pre to Post	37%	43%	35.87%	6%	15%	8.35%
	42%	48%	36.80%	8%	17%	11.32%
Pre to Post*	66%	69%	47.81%	33%	40%	22.81%
	66%	69%	48.34%	30%	37%	28.46%

Note: The term λ_{perm}^{pre} denotes the degree of partial insurance against permanent shocks. * indicates that the cyclicalty of the permanent shocks is shut down. See text for details on the scenarios. The CEV columns denote the corresponding consumption equivalent variation associated with the change from the world with the pre-government income stream and no partial insurance to a world with the pre-government income stream and partial insurance of the size given by λ_{perm}^{pre} .

The obtained partial insurance parameters λ_{perm}^{pre} now combine both, the partial insurance provided by the tax and transfer system, and the additional partial insurance that comes from other insurance channels. Therefore, of course, the obtained λ_{perm}^{pre} reported in Table 3 are larger than the ones reported in the benchmark exercise of Table 2. In order to back out the degree of partial insurance that is provided by taxes and transfers, note that $(1 - \lambda_{perm}^{post})$ is exactly equal to the ratio between the variance of (log) consumption growth and the variance of (log) permanent shocks—see equation (14). Thus, we scale up the consumption variance obtained under pre-government income with partial insurance λ_{perm}^{pre} accordingly and obtain

the government-provided insurance as

$$\lambda^{gov} = 1 - \frac{1 - \lambda_{perm}^{pre}}{1 - \lambda_{perm}^{post}} = \frac{\lambda_{perm}^{pre} - \lambda_{perm}^{post}}{1 - \lambda_{perm}^{post}} \quad (21)$$

Note that for $\lambda_{perm}^{post} = 0$ (our benchmark case), equation (21) implies that $\lambda^{gov} = \lambda_{perm}^{pre}$. For $\lambda_{perm}^{post} = 0.1$, we show the resulting values for λ^{gov} alongside their λ_{perm}^{pre} counterparts in Table 3. Up to rounding error the obtained measures for partial insurance provided by the tax and transfer system are effectively identical to the ones obtained in the benchmark case.

6 Conclusion

The tax and transfer system partially insures households against individual income risk. We discuss under which assumptions differences between income processes estimated for household gross income and disposable income are informative about the (welfare) value of this partial insurance. Our approach works directly with income processes estimated separately on the two income measures, and does not require the specification (nor estimation) of a tax function. Instead we use an incomplete markets framework that links an estimated income process to consumption. Its key feature is that the degree of partial insurance is directly parameterized: Technically, this allows to solve for the degree of insurance provided by the tax and transfer system as a fixed point. The approach works with standard restrictions on income processes and preferences, and it further enables us to explore the role of higher-order risk for the value assigned to public insurance.

Through the lens of our structural model, the degree of overall insurance amounts to 43%, corresponding to 14% in consumption-equivalent terms under log-utility in Sweden. After isolating the gains from a lower initial variance at age 25, the degree of partial insurance amounts to 6% (CEV of about 1.3%). However, the remaining risk in post-government household-level income is still substantial. If cyclical variation of risk was completely eliminated, the partial insurance value would amount to 64%, or a CEV of 16.5%—and thus individuals would be better off by about 3 percentage points of consumption equivalent variation. While the partial insurance value of public insurance is very similar against *skewed*

and *symmetric* income risk, the corresponding CEV gain would be overstated (3%—more than twice as large—against the cyclical component) if the pro-cyclicality in skewness of idiosyncratic risk is ignored.

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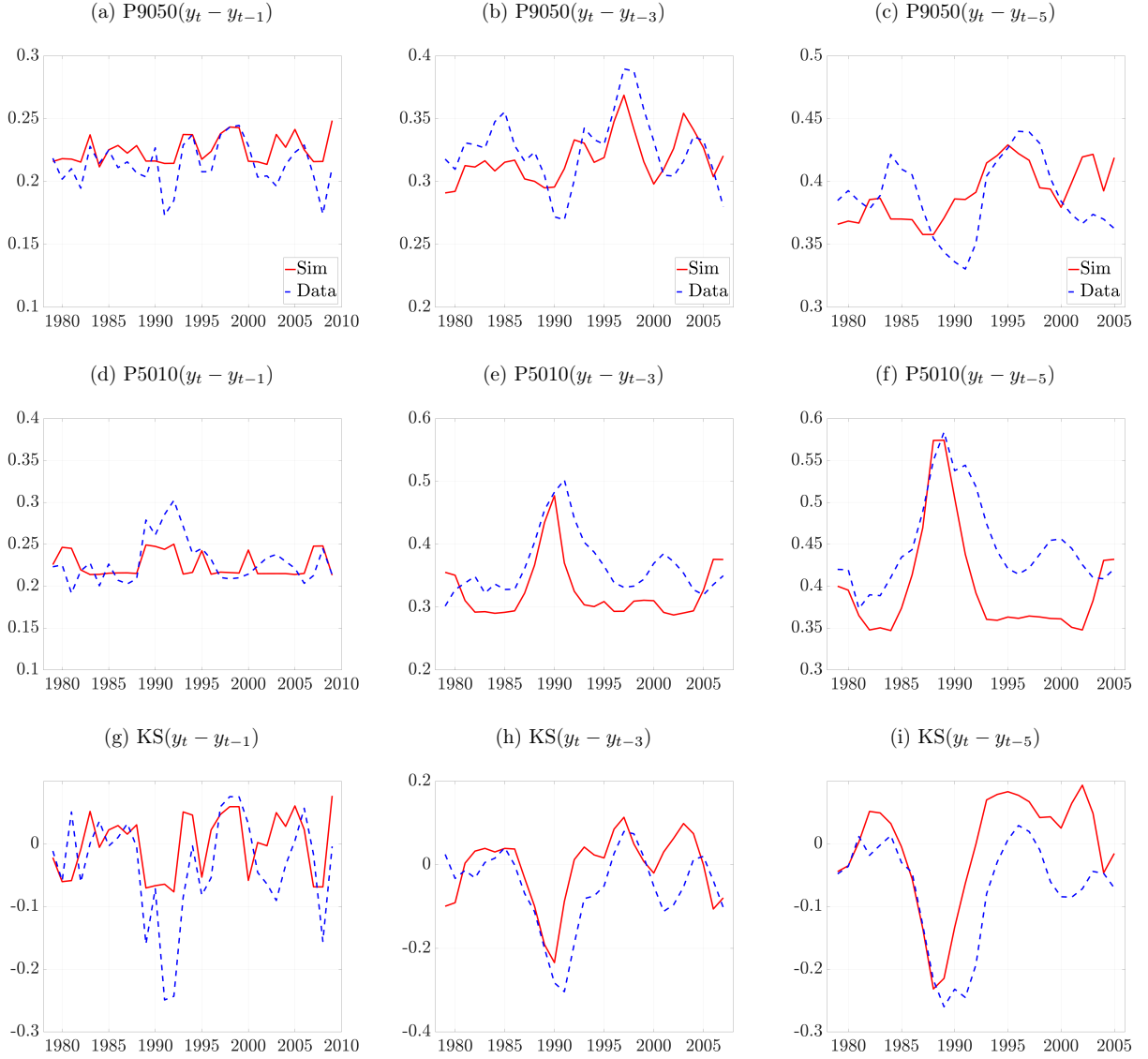
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A Estimated Income Processes

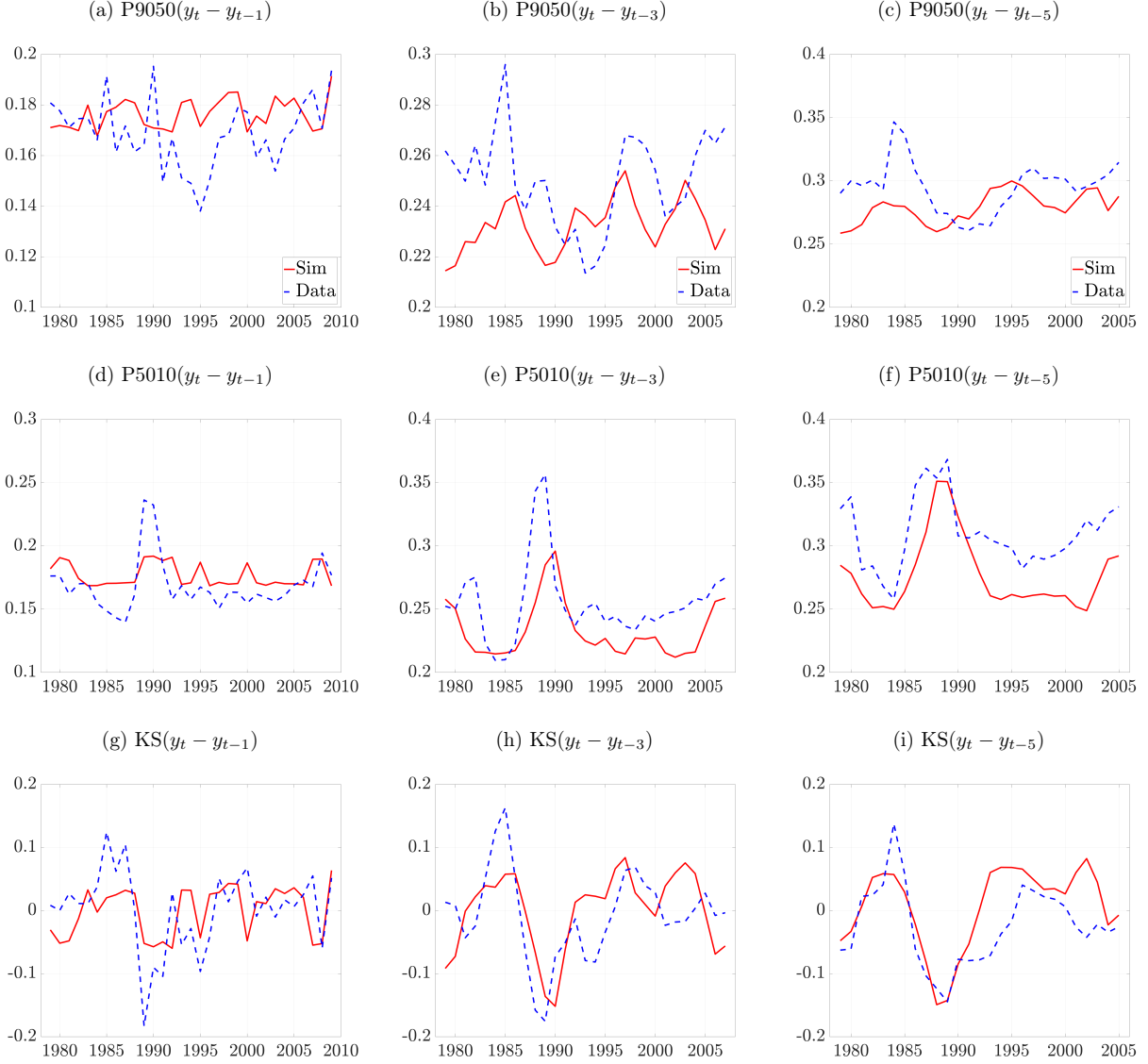
The figures show the estimated income processes for pre- and post-government household income along with the data counterparts of the targeted set of moments.

Figure 3: Pre-Government Income Fit



Note: Each panel shows the time series of a moment of short-run, medium-run, or long-run income changes together with the corresponding moment implied by the estimated income process.

Figure 4: Post-Government Income Fit



Note: See notes to Figure 3.

B Scaling Income Processes

Given estimates of the income process, we scale the parameters of the permanent shocks η to feed them into the model; fraction λ is insurable and the rest is uninsurable. This scaling implies that the first three standardized moments of the distribution of insurable shocks are given as below: for the first three moments of the uninsurable shocks, simply replace λ with $1 - \lambda$.

$$\begin{aligned}
E [\eta_t^{idio}] &= \sum_{i=1}^3 p_{\eta,i} \mu_{\eta^{idio},i,t} = \sum_{i=1}^3 p_{\eta,i} \lambda^{1/2} \mu_{\eta,i,t} = \lambda^{1/2} \sum_{i=1}^3 p_{\eta,i} \mu_{\eta,i,t} = \lambda^{1/2} E [\eta_t] \equiv \lambda^{1/2} \mu_{\eta,t} \\
var [\eta_t^{idio}] &= \sum_{i=1}^3 p_{\eta,i} (\sigma_{\eta^{idio},i}^2 + \mu_{\eta^{idio},i,t}^2) - (E [\eta_t^{idio}])^2 = \sum_{i=1}^3 p_{\eta,i} (\lambda \sigma_{\eta,i}^2 + \lambda \mu_{\eta,i,t}^2) - (\lambda^{1/2} E [\eta_t])^2 \\
&= \lambda \left(\sum_{i=1}^3 p_{\eta,i} (\sigma_{\eta,i}^2 + \mu_{\eta,i,t}^2) - E [\eta_t]^2 \right) = \lambda var [\eta_t] \\
skew [\eta_t^{idio}] &= \frac{1}{var [\eta_t^{idio}]^{3/2}} \sum_{i=1}^3 p_{\eta,i} (\mu_{\eta^{idio},i,t} - E [\eta_t^{idio}]) \left[3\sigma_{\eta^{idio},i}^2 + (\mu_{\eta^{idio},i,t} - E [\eta_t^{idio}])^2 \right] \\
&= \frac{1}{\lambda^{3/2} var [\eta_t]^{3/2}} \sum_{i=1}^3 p_{\eta,i} (\lambda^{1/2} \mu_{\eta,i,t} - \lambda^{1/2} E [\eta_t]) \left[3\lambda \sigma_{\eta,i}^2 + (\lambda^{1/2} \mu_{\eta,i,t} - \lambda^{1/2} E [\eta_t])^2 \right] \\
&= \frac{1}{\lambda^{3/2} var [\eta_t]^{3/2}} \sum_{i=1}^3 p_{\eta,i} \lambda^{1/2} (\mu_{\eta,i,t} - E [\eta_t]) \left[\lambda (3\sigma_{\eta,i}^2 + (\mu_{\eta,i,t} - E [\eta_t])^2) \right] \\
&= \frac{1}{var [\eta_t]^{3/2}} \sum_{i=1}^3 p_{\eta,i} (\mu_{\eta,i,t} - E [\eta_t]) \left[(3\sigma_{\eta,i}^2 + (\mu_{\eta,i,t} - E [\eta_t])^2) \right] \\
&= skew [\eta_t]
\end{aligned}$$